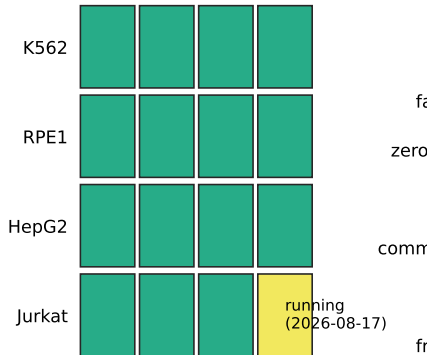


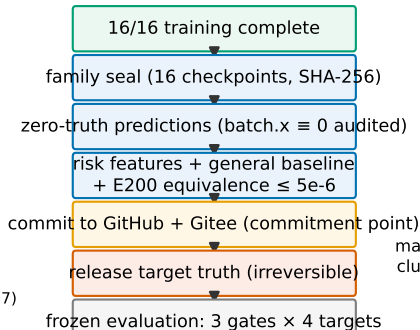
a E201 blind training grid

(leave-one-cell-line-out × 4 seeds)



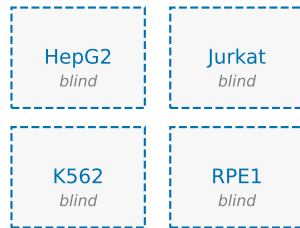
each cell = 80-epoch TxPerf STRING-GAT retraining;
target perturbed expression accessed 0 rows

b Sealed release pipeline (frozen 2026-08-17) until pipeline completes



c Four-target adjudication —

(frozen 2026-08-17) until pipeline completes



main analysis: 1,808 tasks (≥ 30 cells) of 2,008 total;
cluster bootstrap 5,000 × by perturbation condition;
negative results will be reported in full